The Four Phases of Monte Carlo Tree Search

Monte Carlo Tree Search (MCTS) is a simulation-based search algorithm used for intelligent decision making in sequential environments such as Snake.

It operates through four repeated phases.

1. Selection

The algorithm begins at the root node, representing the current Snake game state.

It repeatedly selects the most promising child node using the Upper Confidence Bound for Trees (UCT) formula.

Selection balances:

- Exploitation of known strong moves
- Exploration of less-visited alternatives

The goal is to identify the most promising branch for deeper analysis.

2. Expansion

When the selected node contains unexplored legal moves, one move is chosen and added as a new child node.

Each child represents a possible future Snake state.

The goal is to extend the search frontier.

3. Simulation (Rollout)

From the expanded node, the algorithm simulates future gameplay.

Moves are selected according to the rollout policy until termination or until a rollout limit is reached.

In this project, simulation uses a random rollout policy.

The purpose is to estimate the long-term reward potential of the selected action.

This phase answers:

“If this move is played, how likely is it to produce higher survival and food acquisition?”

4. Backpropagation

The simulation result is propagated back through all visited nodes.

Each node updates:

- Visit count
- Accumulated reward

This information improves future move selection.

Moves that consistently lead to better outcomes gain stronger statistical preference.

Final Decision

After many iterations, the move with the strongest statistics is selected.

Through repeated execution of these four phases, MCTS enables intelligent autonomous gameplay without exhaustive search.